

PH3144 – ELECTRONICS (LAB)
INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH (IISER), PUNE

Project Lumos-Nox:
Bidirectional People Counter &
Automated Energy Management System
for Lecture Halls

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Executive Summary

This report presents the design and implementation of **Project Lumos-Nox**, a smart bidirectional occupancy-sensing and automated energy management system for IISER Pune lecture halls. The system uses a dual-beam infrared sensor gate at the doorway to track the direction of passage (entry vs. exit) and maintain a real-time head-count, driving an automated relay that controls room lighting and fans. A Finite State Machine (FSM) algorithm with timeout logic handles edge cases including partial crossings and group entry. A PIR motion sensor provides a fail-safe override preventing lights from switching off while students sit quietly. A manual override button accommodates instructor-driven control. The system is implemented on an Arduino UNO R3 with an estimated total additional component cost of under **Rs. 500**.

1 Problem Statement

In large lecture halls such as **LHC-101** and common study rooms at IISER Pune, lights, fans, and projectors are frequently left switched on after classes end or during extended inter-class breaks when the room is completely empty. This results in unnecessary standby electricity consumption across the campus throughout the day [6].

Existing partial solutions suffer from well-known limitations:

- **Standard PIR motion sensors:** These fail in examination or lecture settings because students sitting still cause the PIR sensor to time out and switch lights off.
- **Single-beam IR sensors:** These cannot distinguish the direction of crossing (entry vs. exit), making net occupancy determination impossible.
- **Manual switch discipline:** Reliance on the last person leaving to manually switch off is unreliable in practice.

The core problem demands a system that:

1. Tracks *net room occupancy* in real-time.
2. Handles edge cases: partial crossings, group entries, static occupants.
3. Provides a clean, non-intrusive interface installable without rewiring the building.

2 Proposed Solution

Project Lumos-Nox is an automated bidirectional room occupancy counting and energy management system [1]. The name is drawn from the Harry Potter universe: *Lumos* conjures light; *Nox* extinguishes it — precisely mirroring the system behaviour.

The solution comprises four integrated subsystems:

1. **Bidirectional dual-beam gate:** Two IR sensors at the entrance separated by a gap $d \approx 15\text{--}20\text{ cm}$. The order of beam interruption unambiguously identifies direction of crossing [4].
2. **Finite State Machine (FSM) counting algorithm:** A software state machine resolves direction from sensor trigger sequence, applies a timeout to dismiss partial crossings, and maintains a floor-bounded count $N \geq 0$.
3. **Automated relay control:** A 5V relay connected in series with the hall's main distribution board opens when $N = 0$ (Nox) and closes when $N > 0$ (Lumos) [7].
4. **Fail-safe PIR override & manual control:** An interior PIR sensor verifies residual occupancy before power is cut. A manual override button allows instructors to force the system locked-on or locked-off.

3 Theoretical Background

3.1 Infrared Obstacle Detection

Each IR sensor module emits near-infrared radiation ($\lambda \approx 850\text{--}950\text{ nm}$) from an IR LED and detects reflected radiation via a co-located photo-transistor [4]. When an opaque object enters the beam, reflected intensity exceeds a threshold, pulling DO LOW:

$$V_{\text{out}} = \begin{cases} \text{LOW} & I_{\text{reflected}} \geq I_{\text{threshold}} \\ \text{HIGH} & I_{\text{reflected}} < I_{\text{threshold}} \end{cases} \quad (3.1)$$

3.2 Directionality from Temporal Sequencing

Two parallel sensor axes (**S1**: outer corridor side; **S2**: inner hall side) separated by gap d encode direction into trigger ordering. For a person moving at velocity v :

$$\Delta t = \frac{d}{v} \quad (3.2)$$

The count increment is determined by:

$$\Delta N = \begin{cases} +1 & t_{S1} < t_{S2} \quad (\text{entry}) \\ -1 & t_{S2} < t_{S1} \quad (\text{exit}) \end{cases} \quad (3.3)$$

A timeout $\tau \approx 1.5\text{ s}$ guards against partial crossings: if the second sensor is not triggered within τ of the first, the state machine resets without modifying N [1].

3.3 Relay as Electronic Switch

The relay is a single-pole double-throw (SPDT) electromagnetic switch. When the Arduino drives the coil HIGH, the normally open (NO) terminal closes [7]. The mains load is wired through the NO terminal:

$$\text{Load} = \begin{cases} \text{ON} & N > 0 \implies \text{coil energized} \\ \text{OFF} & N = 0 \implies \text{coil de-energized} \end{cases} \quad (3.4)$$

The relay provides complete galvanic isolation between the 5 V DC Arduino control circuit and the 230 V AC mains load [7, 3].

4 System Architecture

4.1 Hardware Layers

1. **Sensing:** Dual IR gate + PIR interior sensor.
2. **Processing:** Arduino UNO R3 (ATmega328P microcontroller) running the FSM algorithm [2, 3].
3. **Actuation:** 5V Relay + $16 \times 2\text{ I}^2\text{C LCD}$ (LiquidCrystal_I2C) + Manual override button [5].

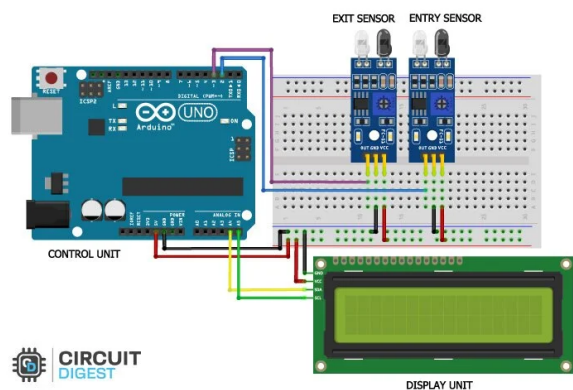


Figure 1: Phase 1 circuit architecture: Preliminary reference schematic featuring the dual-beam IR obstacle sensor gate with a 16×2 I²C LCD backpack (interfacing via A4/A5).

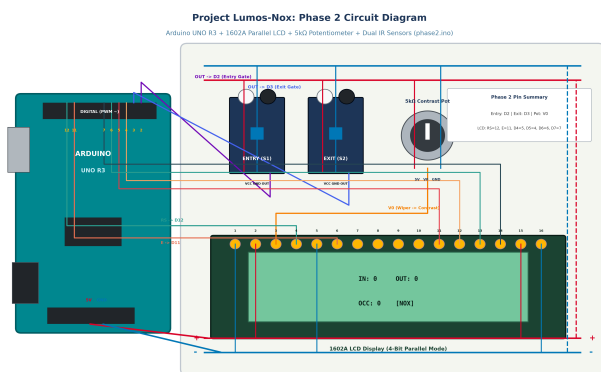


Figure 2: Phase 2 physical breadboard circuit diagram corresponding to `phase2.ino`: Implements the dual active IR obstacle sensors (Entry on D2, Exit on D3), a 5 kΩ contrast potentiometer, and the 1602A LCD operating in 4-bit parallel mode (RS=12, E=11, D4=5, D5=4, D6=6, D7=7).

4.2 Hardware Circuit Schematics & Pin Connections

Table 1: Physical wiring summary corresponding to `phase2.ino`.

Component	Pin Label	Arduino / Rail
IR Sensor 1 (Entry)	OUT	Digital 2
IR Sensor 2 (Exit)	OUT	Digital 3
Both IR Sensors	VCC, GND	5V Rail, GND Rail
1602A LCD	VSS (Pin 1)	GND Rail
1602A LCD	VDD (Pin 2)	5V Rail
1602A LCD	V0 (Pin 3)	5kΩ Pot Wiper (Middle)
1602A LCD	RS (Pin 4)	Digital 12
1602A LCD	RW (Pin 5)	GND Rail
1602A LCD	E (Pin 6)	Digital 11
1602A LCD	D4 (Pin 11)	Digital 5
1602A LCD	D5 (Pin 12)	Digital 4
1602A LCD	D6 (Pin 13)	Digital 6
1602A LCD	D7 (Pin 14)	Digital 7
1602A LCD	A (Pin 15)	5V Rail
1602A LCD	K (Pin 16)	GND Rail
5kΩ Potentiometer	Outer Legs	5V Rail, GND Rail
5kΩ Potentiometer	Middle Leg	LCD V0 (Pin 3)

5 Algorithm Design

5.1 Finite State Machine

The counting logic is implemented as a timeout-governed state engine:

1. **IDLE:** Both sensors clear, awaiting first beam break.

2. **ENTRY SEQUENCE:** Sensor 1 (Entry) triggers LOW. System opens a time window ($\tau \approx 1000$ ms) using `millis()`. If Sensor 2 triggers within τ , `in_count` is incremented by +1.
3. **EXIT SEQUENCE:** Sensor 2 (Exit) triggers LOW. System opens timeout window. If Sensor 1 triggers within τ , `out_count` is incremented by +1 (subject to $N_{out} \leq N_{in}$).
4. **STATE CLEARING:** The system blocks until both sensors return to an unblocked state to prevent multiple triggers from a single passage.

Net occupancy is computed dynamically:

$$N_{\text{current}} = \max(N_{\text{in}} - N_{\text{out}}, 0) \quad (5.1)$$

5.2 PIR Safety Override

Before cutting power at $N_{\text{current}} = 0$, the system queries the PIR over a 10-second verification window. If residual motion is detected, power is maintained.

5.3 Manual Override Modes

- **AUTO:** Relay follows N_{current} .
- **FORCE ON:** Relay held closed regardless.
- **FORCE OFF:** Relay held open regardless.

6 Implementation Phases & Progress

Table 2: Project development roadmap and verification status.

Ph.	Goal	Deliverable	Status
1	People counter	Dual IR + 4-bit 1602A LCD live count	Completed
2	Lumos/Nox Engine	Automated state transitions & display	Completed
3	Override + PIR	Full FSM + safety + 3-mode button	Next up
4	Enclosure & Demo	Scale prototype, final calibration	Planned

Both Phase 1 and Phase 2 have been physically constructed, calibrated, and empirically validated on the bench with the Arduino UNO, dual active IR obstacle sensors, 5kΩ potentiometer contrast tuning, and the 1602A LCD display.

7 Firmware Implementations

7.1 Hardware Diagnostic & Calibration Routine

To calibrate the detection thresholds of both IR sensors and eliminate comparator false triggers from ambient lighting, the following diagnostic routine (`test_sensors.ino`) was deployed. It provides simultaneous visual verification on the 1602A LCD and real-time digital logic telemetry via the Serial Monitor at 9600 baud.

```

1 #include <LiquidCrystal.h>
2
3 LiquidCrystal lcd(12, 11, 5, 4, 6, 7);
4
5 const int irPin1 = 2; // Outer Sensor
6 const int irPin2 = 3; // Inner Sensor
7
8 void setup() {
9   Serial.begin(9600);
10  lcd.begin(16, 2);
11  pinMode(irPin1, INPUT_PULLUP);
12  pinMode(irPin2, INPUT_PULLUP);
13 }
```

```

14
15 void loop() {
16   int s1 = digitalRead(irPin1);
17   int s2 = digitalRead(irPin2);
18
19   // Serial Monitor Telemetry
20   Serial.print("S1:"); Serial.print(s1);
21   Serial.print("_|_S2:"); Serial.println(s2);
22
23   // Real-time LCD Diagnostic Output
24   lcd.setCursor(0, 0);
25   lcd.print("S1 (Pin2):");
26   lcd.print(s1 == LOW ? "DETECT_" : "CLEAR_");
27
28   lcd.setCursor(0, 1);
29   lcd.print("S2 (Pin3):");
30   lcd.print(s2 == LOW ? "DETECT_" : "CLEAR_");
31
32   delay(150);
33 }

```

Listing 1: Hardware Diagnostic & Calibration Tool (test_sensors.ino)

7.2 Phase 2 Production Firmware

The complete Phase 2 production sketch (phase2.ino) implements the non-blocking temporal timeout engine ($\tau = 1200$ ms), direction parsing, non-negative occupancy constraint, splash initialization, and formatted 1602A LCD UI.

```

1  #include <LiquidCrystal.h>
2
3  // RS=12, E=11, D4=5, D5=4, D6=6, D7=7
4  LiquidCrystal lcd(12, 11, 5, 4, 6, 7);
5
6  const int irPin1 = 2; // Outer Entrance Gate
7  const int irPin2 = 3; // Inner Exit Gate
8
9  int in_count = 0;
10 int out_count = 0;
11 int current_count = 0;
12
13 const unsigned long timeout = 1200; // 1.2s timeout
14   window
15
16 void updateDisplay();
17
18 void setup() {
19   Serial.begin(9600);
20   lcd.begin(16, 2);
21   lcd.clear();
22
23   // Welcome Splash Screen
24   lcd.setCursor(0, 0);
25   lcd.print("_PROJECT_LUMOS_");
26   lcd.setCursor(0, 1);
27   lcd.print("_SYSTEM_READY_");
28   delay(1500);
29   lcd.clear();
30
31   pinMode(irPin1, INPUT_PULLUP);
32   pinMode(irPin2, INPUT_PULLUP);
33
34   updateDisplay();
35   Serial.println("===_Project_Lumos-Nox:_Phase_2_Active_");
36
37 void loop() {
38   // Entry: S1 then S2
39   if (digitalRead(irPin1) == LOW) {
40     unsigned long startTime = millis();
41     while ((millis() - startTime) < timeout) {
42       if (digitalRead(irPin2) == LOW) {

```

```

43       ++in_count;
44       Serial.print(">>_[ENTRY]_Total_IN:");
45       Serial.print(in_count);
46       Serial.print("_|_Current:");
47       Serial.println(in_count - out_count);
48       updateDisplay();
49       break;
50     }
51   }
52   while (!digitalRead(irPin1) || !digitalRead(irPin2))
53     delay(50);
54 }
55
56 // Exit: S2 then S1
57 else if (digitalRead(irPin2) == LOW) {
58   unsigned long startTime = millis();
59   while ((millis() - startTime) < timeout) {
60     if (digitalRead(irPin1) == LOW) {
61       if (out_count < in_count) {
62         ++out_count;
63         Serial.print("<<_[EXIT]_Total_OUT:");
64         Serial.print(out_count);
65         Serial.print("_|_Current:");
66         Serial.println(in_count - out_count);
67         updateDisplay();
68         break;
69       }
70     }
71   }
72   while (!digitalRead(irPin1) || !digitalRead(irPin2))
73     delay(50);
74 }
75
76 void updateDisplay() {
77   current_count = in_count - out_count;
78   if (current_count < 0) current_count = 0;
79
80   // Row 0: "IN: xx  OUT: yy"
81   lcd.setCursor(0, 0);
82   lcd.print("IN:");
83   lcd.print(in_count);
84   lcd.print("_|_");
85
86   lcd.setCursor(9, 0);
87   lcd.print("OUT:");
88   lcd.print(out_count);
89   lcd.print("_|_");
90
91   // Row 1: "OCC: zz [LUMOS]" or "OCC: 0 [NOX]"
92   lcd.setCursor(0, 1);
93   lcd.print("OCC:");
94   lcd.print(current_count);
95   lcd.print("_|_");
96
97   lcd.setCursor(8, 1);
98   if (current_count == 0) {
99     lcd.print("_[NOX]_");
100   } else {
101     lcd.print("_[LUMOS]_");
102   }
103 }
104 }

```

Listing 2: Phase 2 – Operational Lumos-Nox Firmware (phase2.ino)

8 Bill of Materials

Table 3: Component list and cost estimate.

Component	Q	Source	Cost
Arduino UNO R3	1	UG Lab	Free
IR Sensor Module	2	37-in-1 Kit	Free
PIR Motion Sensor	1	37-in-1 Kit	Free
5V Relay Module	1	37-in-1 Kit	Free
LDR Light Sensor	1	37-in-1 Kit	Free
Piezo Buzzer	1	37-in-1 Kit	Free
16×2 I ² C LCD	1	Robu.in	Rs. 180
Pushbutton	1	37-in-1 Kit	Free
Breadboard + wires	–	UG Lab	Free
Total			Rs. 180

9 Conclusion

Project Lumos-Nox addresses a concrete energy waste problem at IISER Pune lecture facilities. By combining directional infrared sensing, a robust FSM counting algorithm, relay-driven mains control, and a PIR safety override, the system reliably automates room lighting management without modifying existing manual switches or building infrastructure.

The phased implementation plan validates the core counting logic first, before integrating relay and override subsystems. The total hardware expenditure is under **Rs. 500**, well within the course-allocated budget of Rs. 2,000 per group. Beyond the engineering deliverable, the project demonstrates core electronics principles including infrared photodetection, relay switching, I²C communication, and embedded FSM design — directly corresponding to the learning objectives of PH3144.

References

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